AZINHACK '26 · TEAM RED-HAT

PARDARSHI पारदर्शी

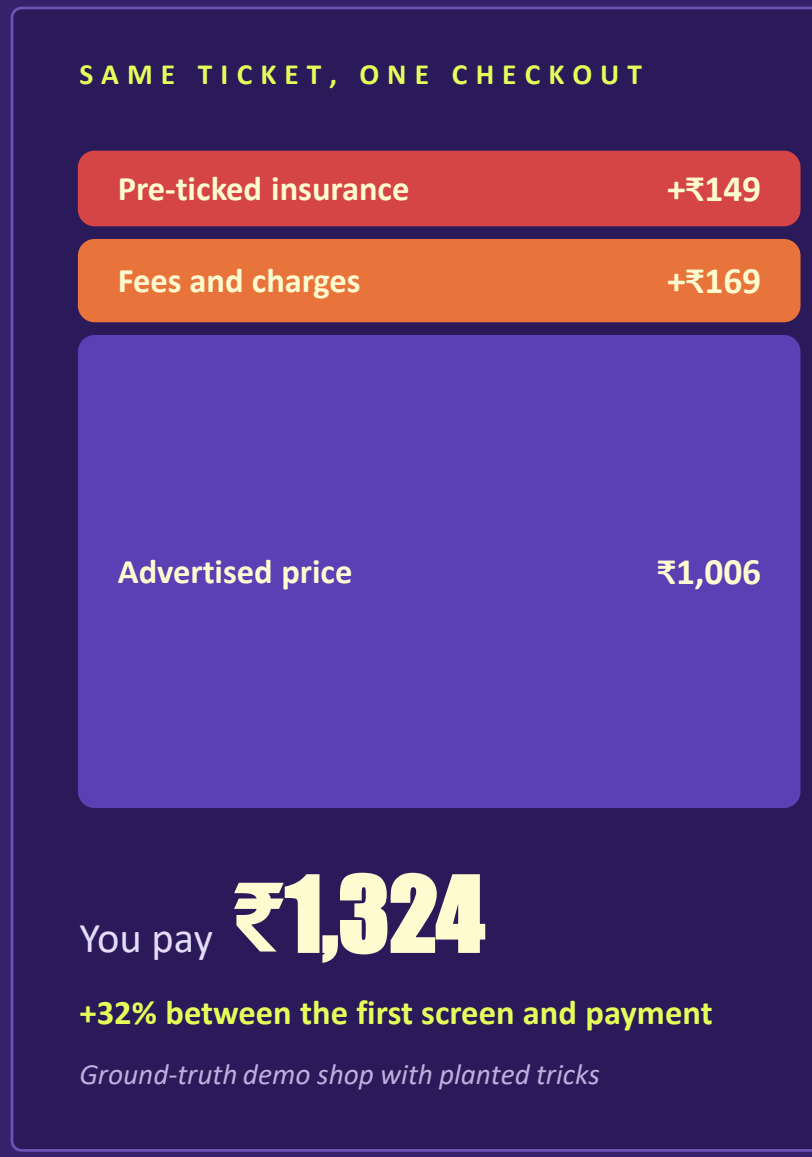
AI shopper agents go through checkout like real people, stop before paying, and measure the hidden fees, fake urgency and personalised prices.

An independent consumer-protection auditor for India's online shoppers

Working product | Live on Indian sites | TinyFish on a live shop | 94 tests

Built with TinyFish · Google ADK · Gemini · scikit-learn · Playwright

Demo video https://youtu.be/xR9-eA1mp7o · **Code** https://github.com/mathurojus/Pardarshi



AT A GLANCE

Every section the judges score, answered in one line, with the slide that proves it

Section	Answer	Slide
Problem Statement	The price shown first is not the price paid; one shopper cannot see fees, pre-ticks or personalised prices.	**3**
Proposed Solution	AI shopper agents run checkout as 6 profiles, stop before payment and measure 7 dark patterns as evidence.	**4–6**
Target Users	Consumer-rights investigators and journalists; platform compliance teams; shoppers and regulators.	**7**
Role of ML and AI	Trained copy classifier (macro-F1 0.89 on unseen text), price regression with BH correction; Gemini agent + ADK.	**8–9**
Technical Approach	FastAPI pipeline, two browser lanes, evidence at every step, hard pay-button guard, 94 tests.	**10–11**
Real-World Validation	Audited live Indian shops: bummer.in and fardaclothing.com reached payment; 30 sites probed.	**12**
TinyFish Integration	All 4 APIs wired and verified; Agent API walked bummer.in to checkout in 13 steps.	**13–14**
Expected Impact	Suspicion to statistics, complaint to case file; pilot: 2 NGOs, 50 Indian sites, about ₹22,000.	**15**
Feasibility	Working today; honest risk list with a mitigation for each (bot walls reported, never bypassed).	**16**
Future Scope	India exit nodes, Indic-language classifier, shopper capture extension, NCH filing.	**17**

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PROBLEM STATEMENT

Online prices change between the product page and payment, where a single shopper cannot see it

₹25–28k cr

lost by Indian online buyers every year to dark patterns and hidden charges

97%

of 290 major Indian platforms use at least one dark pattern

304M

online buyers in India; 88% lose ₹78–87 a month to hidden charges

4.2 lakh

e-commerce complaints a year at the National Consumer Helpline, 32% of all

Why it persists

Fees appear only at checkout

Drip pricing and pre-ticked add-ons wait on the last screen, after the shopper has committed.

Personalised prices are invisible

One shopper only ever sees their own price, so device or location pricing goes unnoticed.

Compliance is self-declared

The CCPA asked 50+ platforms to self-audit in June 2025. No independent check is required.

Sources: ₹25–28k cr, 304M: Datum Intelligence (Jun 2026) · 97%: LocalCircles (Oct 2025) · 4.2 lakh: NCH data FY2024-25 · 50+ platforms: CCPA advisory, 5 Jun 2025 (as reported).

PROPOSED SOLUTION

Pardarshi measures what shoppers are actually charged, and turns it into evidence

01

Shop

6 shopper profiles × 3 runs, plus the cancellation flow

02

Stop

Every agent stops before payment; a code guard blocks pay buttons

03

Record

Every screen: price lines, pre-ticked boxes, urgency claims, screenshots

04

Measure

ML and statistics turn captures into findings with confidence

05

Act

Trust score, findings report with SHA-256 hashes, NCH complaint

Measured, not surveyed. Per-site proof with screenshots and statistics, ready to attach to a complaint.

PROPOSED SOLUTION: SEVEN CHECKS

Mapped to the Consumer Protection Act 2019 and the CCPA's 2023 dark-pattern guidelines



Drip pricing

First-screen price vs payment total, median of runs

CCPA 2023: drip pricing



Basket sneaking

Paid add-ons that arrive pre-ticked

CCPA 2023: basket sneaking



Fake urgency

Timers that reset on reload, “only N left”, fake popularity

CCPA 2023: false urgency



Manipulative copy

Confirm-shaming and nagging copy, by the ML classifier

CCPA 2023: confirm shaming, nagging



Personalised pricing

Same product, price by device, country or visitor

CPA 2019: unfair trade practice



Cancellation trap

Clicks to cancel against clicks to join

CCPA 2023: subscription trap



Policy vs reality

Written promises the measurements disprove

CPA 2019: misleading advertisement

Every finding carries

its legal category, severity, penalty points, the run it came from and its screenshots.

PROPOSED SOLUTION: WHAT IS NEW

Research proved it can be measured; Pardarshi makes it a repeatable, complaint-ready audit for India

Approach	Real charge	Compares profiles	Repeatable + stats	Independent
Consumer surveys (e.g. LocalCircles)	Self-reported	**No**	**No**	**Yes**
Studies (Hannak '14, Mathur '19)	**Yes**	**Yes**	One-off studies	**Yes**
Detectors and browser extensions	One page, one shopper	**No**	**No**	**Yes**
Manual investigation	**Yes**	Rarely	**No**	**Yes**
Platform self-audit (CCPA, 2025)	Not disclosed	Not disclosed	Not disclosed	**No**
Pardarshi	**Yes, at checkout**	**Yes, 6 profiles**	**Per site, on demand**	**Yes**

Evidence, not opinion

Screenshots, price lines and SHA-256 hashes behind every finding.

Several shopper profiles

Personalised pricing only shows when profiles are compared.

Ready to act on

Findings map to Indian law and become an NCH complaint.

TARGET USERS AND USE CASE

Built for people who need proof, not opinions

PRIMARY

Consumer-rights investigators and journalists

Today: burner accounts, manual screenshots, a single shopper's view.

With Pardarshi: many shopper runs, statistics and a ready case file.

SECONDARY

Platform compliance teams

Today: the CCPA's 2025 advisory asks for a self-audit and a self-declaration.

With Pardarshi: an independent, repeatable check of their own site, with consent.

BENEFICIARY

Shoppers and regulators

Today: vague complaints (“they overcharged me”) that are hard to act on.

With Pardarshi: complaints that name the category, the figures and the exhibits.

Use case, measured Demo shop: ₹1,006 became ₹1,324 at payment (+32%), iPhone shoppers were quoted 9.6% more, cancelling took 8 clicks against 1. Live: an investigator reaches the payment page of an Indian shop in about 2 minutes.

ROLE OF MACHINE LEARNING

A trained classifier, statistics and rules turn captures into findings, measured on unseen data

Dark-pattern copy classifier (trained)

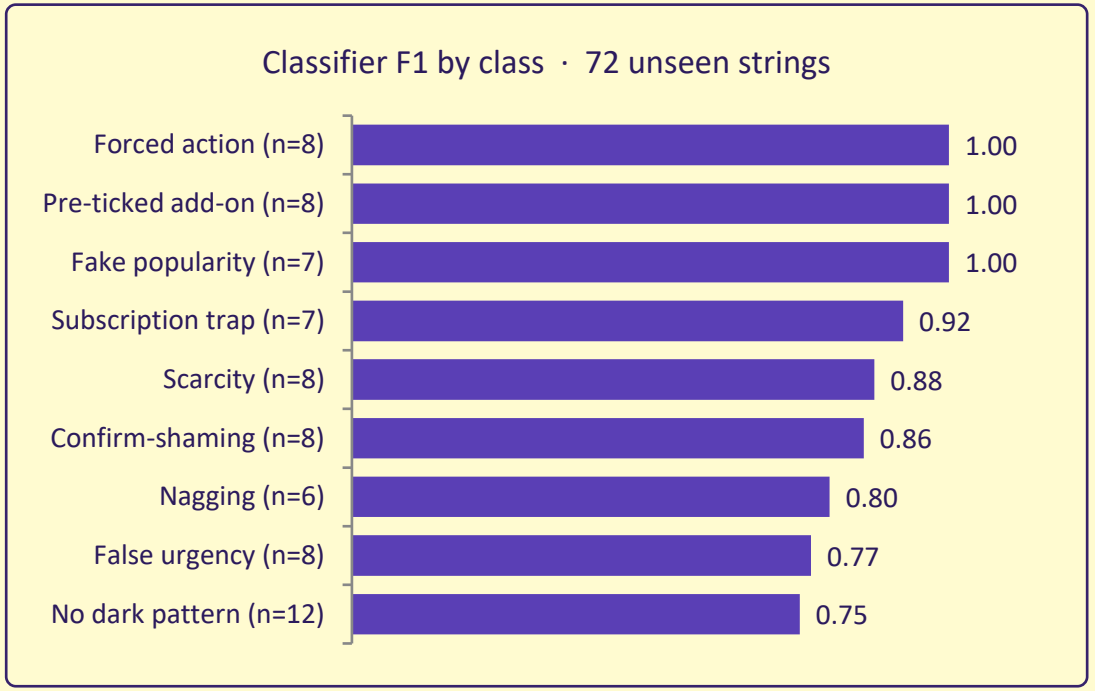
TF-IDF word and character n-grams into logistic regression; 9 classes, 725 labelled UI strings. Tested on strings never seen in training.

Personalised-pricing statistics

Regression on log price by device, country and visitor, 95% confidence intervals, Benjamini–Hochberg correction, 2% minimum gap.

Price extraction and trust score

Page text into labelled lines: item, fee, tax, add-on, discount, total. Score = 100 × e^(−points / 70), each point traced to a finding.



0.89
macro-F1, unseen strings

1.00
precision, dark vs none (recall 0.87)

12/12
planted tricks found (demo shop)

Per-class n is small (6–12), so these are early numbers. On real Indian sites the price extractor was hardened on live checkouts.

ROLE OF AI

Gemini and Google ADK do the reasoning; guards keep the agents from ever paying

Shopper agent (Gemini)

Reads each page and picks one action at a time towards checkout, on sites nobody scripted. Recovers from pop-ups, redirects and busy models.

Guards that stop payment

Code blocks Pay, Confirm order and Subscribe, and never types card, OTP or personal data. Proven live: it refused boAt's “Pay Now”.

Policy vs reality (Gemini)

Reads the site's terms and refund pages and flags written promises that the measured runs disprove.

Google ADK orchestrator

A tool-using agent that runs audits, finds peer sites (TinyFish Search), reads policies (Fetch) and drafts complaints.

Voice and text assistant

Exact answers from the audit's own numbers; open questions go to Gemini with the whole audit as context.

Complaint writer

Drafts a National Consumer Helpline complaint that cites only measured findings, with exhibits.

PROPOSED TECHNICAL APPROACH

One pipeline, two browser providers, evidence captured at every step

Console

FastAPI, live agent view, voice and text assistant

Orchestrator

Google ADK + pipeline: shopper profiles in random rounds, one pinned product

Browsers

TinyFish Browser and Agent API, or Playwright (open source)

Capture

Every screen: prices, copy, checkboxes, screenshots

Analysis

Classifier, price statistics, Gemini policy check

Outputs

Score, findings report, SHA-256 hashes, NCH complaint

Stack

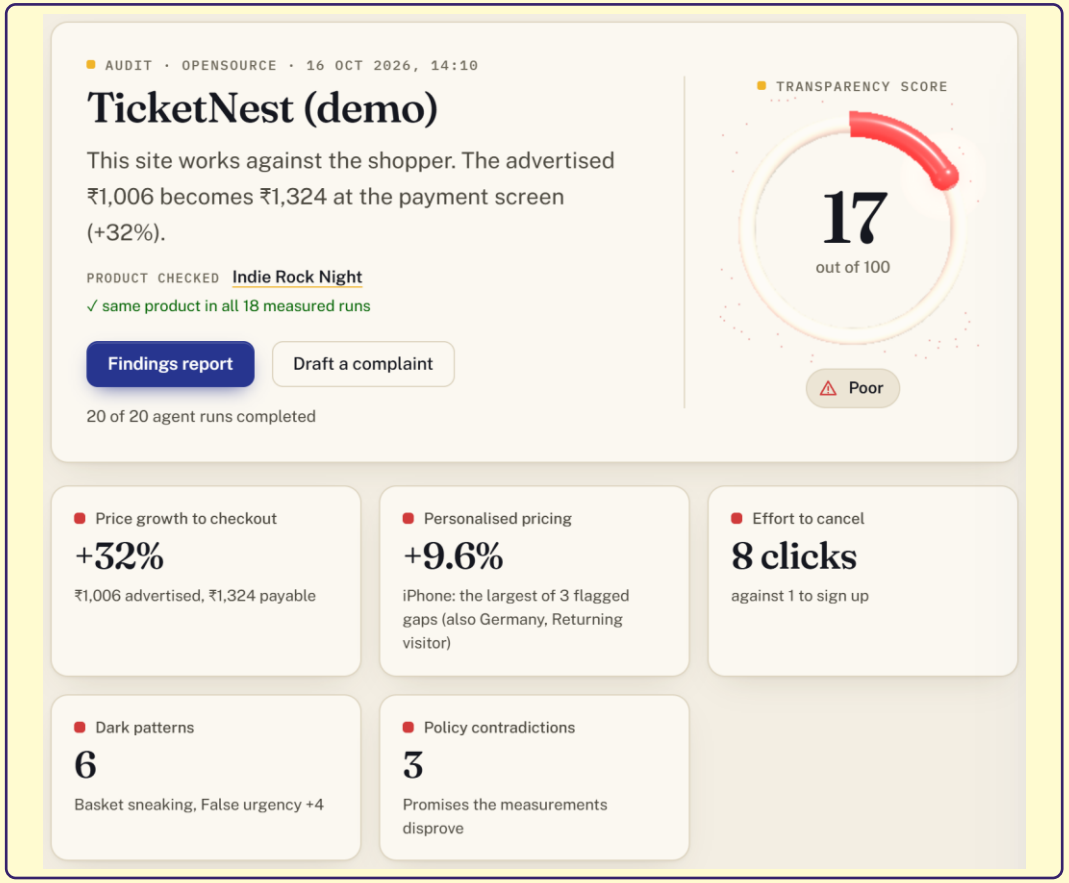
Python · FastAPI · TinyFish SDK · Playwright · Google ADK · Gemini · scikit-learn · SciPy · SQLite

Safety by design

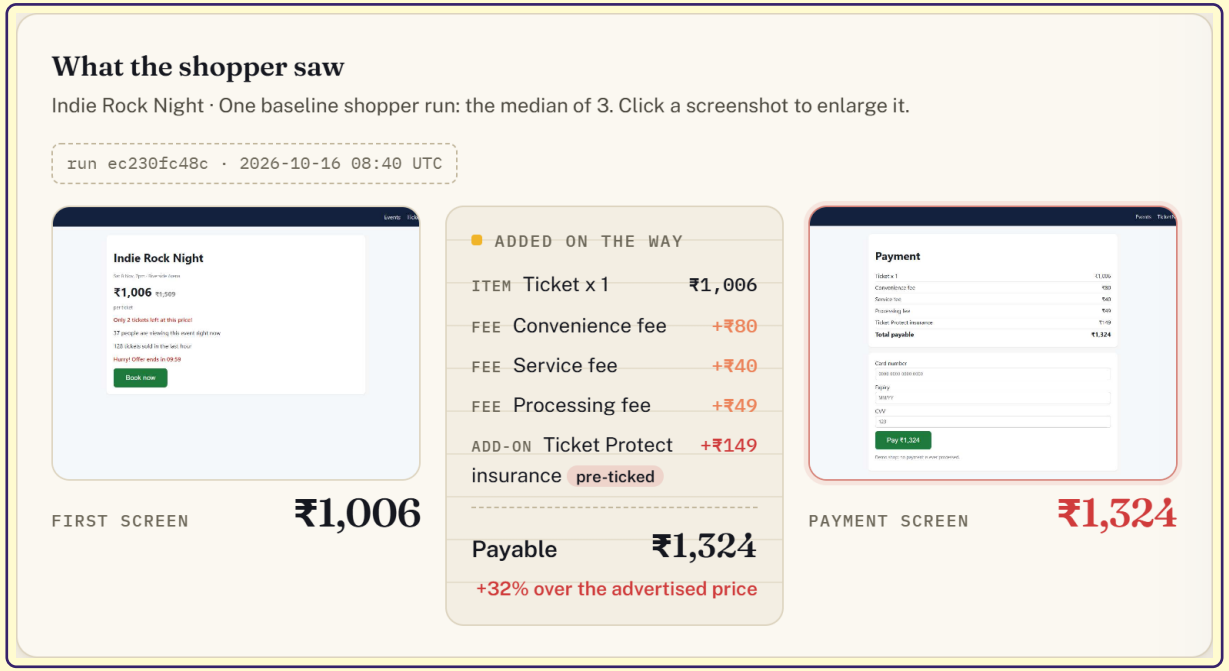
- Allowlisted sites only
- Stops before payment
- Pay-button guard in code
- One shopper at a time
- Blocks reported, not bypassed
- Private, per-user audits

PROPOSED TECHNICAL APPROACH: WORKING PRODUCT

Three real screens from one audit: no mock-ups



Console: score, verdict, same product in all runs, key measurements



Evidence: first screen vs payment, every added line



Findings report
Reference, verdict stamp, exhibits, SHA-256 integrity section

REAL-WORLD VALIDATION: INDIAN SITES

Live Indian shops, audited as Indian shoppers, 16 Oct 2026: what worked and what blocked us

bummer.in — **Payment page · ₹599**

Reached the payment page as an Indian shopper in about 2 minutes; one product pinned, every screen captured.

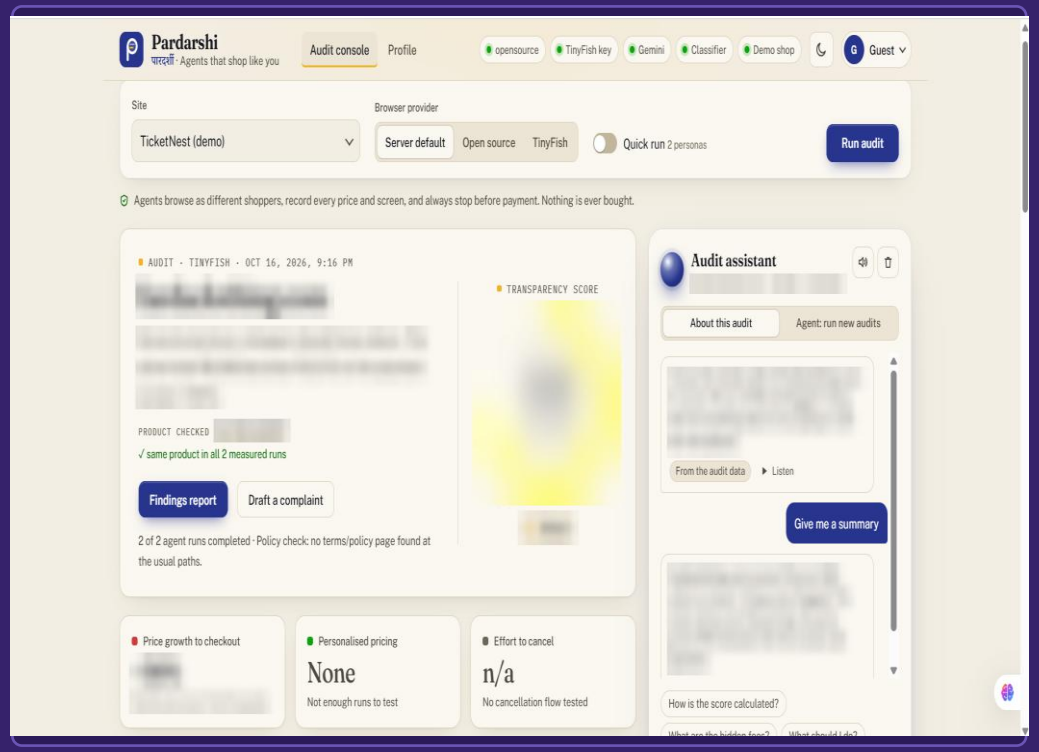
fardaclothing.com — **Payment · ₹6,999**

Reached the payment page in under a minute; checkout read and captured as evidence.

bummer.in · TinyFish — **Checkout · ₹599**

Agent API: product to cart to checkout in 13 steps, stopped before payment, $0.21.

30 Indian sites probed. 8 large platforms (Myntra, Meesho, Ajio, Nykaa, MakeMyTrip, BookMyShow, BigBasket, Croma) block bots at the door: reported, never bypassed. Login walls: use Connect account.



TinyFish audit in the console (shop name and figures withheld)

Hardened on live checkouts: pop-ups and new tabs, mid-load redirects, Shopify totals, busy-model retries.

PROPOSED TINYFISH INTEGRATION

All four TinyFish APIs integrated, each with a specific job, and run live on a real Indian shop

Browser API

per shopper run

A cloud browser per shopper over the Chrome DevTools Protocol (CDP), with a country proxy (US, GB, CA, DE, FR, JP, AU). Our flows set device, locale and timezone, take screenshots and reload to expose fake countdowns.

Agent API

unscripted sites

A goal (reach checkout, never pay) plus a typed output_schema: advertised price, every price line with its pre-ticked flag, verbatim urgency copy, the product and where it stopped.

Fetch API

policy pages

Terms, refund and cancellation pages as clean markdown, up to 10 URLs per call, feeding the Gemini policy-vs-reality check.

Search API

peer sites

Finds comparable services on the live web for the Google ADK agent's peer-site tool, so a site can be compared with its peers.

Live on a real Indian shop: Agent API on bummer.in: 13 agent steps, stopped before payment, ₹599 returned as typed JSON for $0.21.
Cost: that run $0.21 · ≈ $0.04 per 20-run Browser API audit · Search and Fetch $0 (TinyFish rates, 16 Oct 2026)

PROPOSED TINYFISH INTEGRATION: ONE RUN

One shopper run through TinyFish, step by step, with its real output

1. **Pick a shopper**
   e.g. iPhone · Germany · new visitor
2. **Open a session**
   Browser API with a country proxy, or the Agent API
3. **Become that shopper**
   CDP: device metrics, user agent, locale, timezone
4. **Walk the checkout**
   Pinned product → basket → checkout, every screen captured
5. **Stop safely**
   At payment, a login wall or a CAPTCHA (reported, never bypassed)
6. **Hand back evidence**
   Typed JSON + screenshots → RunRecord → ML and statistics

Agent API · real output from bummer.in (abridged)


```
{
  "product_name": "Modal Stretch Trunks",
  "advertised_price": 599,
  "final_payable_total": 599,
  "tax_included": 28.52,
  "stopped_at": "payment",
  "steps": [ ... ]
}
```


Why TinyFish

- **Real geography:** country exit IPs, verified live; India exits are on our roadmap.
- **Unscripted sites:** the Agent API walked a live Indian checkout nobody scripted.
- **Managed browsers:** our effort goes into measurement, not infrastructure.

EXPECTED IMPACT

From a shopper's suspicion to evidence a regulator can act on

One shopper's view → **Many profiles**

6 shopper profiles × 3 runs; real sites one shopper at a time

Suspicion → **Statistics**

price gaps with 95% confidence intervals and a noise floor

Complaint → **Case file**

an NCH-ready letter with exhibits and SHA-256 hashes

Self-declaration → **Independent check**

a repeatable re-check a regulator can ask for

+32%

first screen to payment (demo shop)

+9.6%

quoted to iPhone shoppers, 95% CI 8.4–10.9% (demo)

8 vs 1

clicks to cancel against clicks to join (demo)

13

Agent API steps to a live Indian checkout

Pilot goal **2 consumer NGOs · 50 Indian sites in 3 months · ≈ ₹22,000 on the TinyFish Agent API (≈ ₹440 per audit), then monthly re-checks**

FEASIBILITY

Built, tested on real sites and running today, with an honest list of risks

Built and working

- End to end: 20 of 20 runs in 35 s on the demo shop
- Live Indian shops audited: bummer.in, fardaclothing.com
- TinyFish Agent API run on bummer.in; all 4 APIs verified
- 94 automated tests: ML, flows, guards, auth, TinyFish
- Classifier on unseen strings: macro-F1 0.89
- Open source on GitHub; private accounts, live console, report

Risk	Mitigation
Bot protection	Measured: 8 of 30 sites block. Reported with evidence, never bypassed; standard browsers; own-account session
Login before payment	Measure to the furthest price screen; Connect account (the user signs in in their own Chrome)
False positives	Noise floor, Benjamini–Hochberg, 2% minimum gap, random run order; a person files
Evidence in court	SHA-256 hashes reveal tampering; next: trusted timestamps, Section 63 (BSA 2023) certificate
Legal and ethics	Allowlisted sites, observe only, never pays, one shopper at a time
Cost	₹0 open-source; ≈ ₹3.5 per Browser API audit, ≈ ₹440 per Agent API audit

FUTURE SCOPE

From a hackathon tool to public consumer-protection infrastructure

NEXT 3 MONTHS

- Pilot with consumer NGOs on 50 Indian sites
- India exit nodes for geo-pricing tests
- Hindi and Indic-language copy classifier

6 MONTHS

- Shopper capture extension: real people, no bot
- Continuous monitoring with alerts
- Regulator dashboard across sites

12 MONTHS

- Platform allowlisting for independent audits
- Filing into NCH's INGRAM complaint portal
- Apps and quick-commerce audits

Sustainability free for shoppers and consumer NGOs; paid, consented recurring audits for platforms that want to prove compliance.

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MEASURED, NOT SURVEYED

Pardarshi sends AI shoppers through checkout, stops before paying, and hands consumers and regulators the evidence: the hidden fees, the fake urgency and the personalised prices.

4
TinyFish APIs, run live

7
checks mapped to Indian law

13
agent steps to a live Indian checkout

94
automated tests

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Thank you पारदर्शी